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PenuX-SAP Stage I Observational Cohort Study Protocol

Full Title: Prospective Observational Validation of a Machine-Learning-Based Severity Prediction System (PenuX) in Adults Admitted with Acute Pancreatitis: A Stage I Cohort Study

Protocol Version: 1.0

Date: June 2026

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ClinicalTrials.gov: [Pending Registration]

1. Background and Scientific Rationale

Acute pancreatitis (AP) is one of the most common gastrointestinal emergencies, with a global annual incidence of approximately 34 per 100,000 persons and a rising trend. Roughly 10-20% of patients progress to severe acute pancreatitis (SAP), defined by the 2012 Revised Atlanta Classification (RAC) as persistent organ failure lasting more than 48 hours. SAP carries a mortality of 20-40% and places substantial demands on intensive care resources.

Current severity scoring systems ? Ranson criteria, APACHE II, BISAP, and the Modified CT Severity Index ? require 24-48 hours of observation before reliable stratification is possible, or depend on contrast-enhanced CT imaging that may be unsafe in the acute phase. This delay limits early triage and timely escalation of care.

PenuX is a machine-learning ensemble that predicts SAP severity at the time of hospital admission using routine laboratory values. In a retrospective computational study of 722 AP patients labelled according to the 2012 Revised Atlanta Classification, PenuX (Random Forest core) achieved AUC-ROC 0.877, sensitivity 96.8%, F1 0.917 and PPV 87.1% ? outperforming all deep-learning architectures tested. A secondary pancreatic sepsis risk sub-model provides probabilistic sepsis risk stratification at admission.

This Stage I study is the first prospective external validation of the PenuX system in a real-world hospital population, conducted in silent-mode (prediction-only, no clinical intervention). The outputs of this stage will inform the design of a Stage II prospective interventional pilot.

2. Objectives

2.1 Primary Objective

Assess the external discriminative validity of the PenuX SAP severity prediction model in a prospective cohort of AP admissions, measured by AUC-ROC against the RAC gold-standard severity label at 48 hours.

2.2 Secondary Objectives

1. Evaluate sensitivity, specificity, PPV, NPV, and F1 at the pre-specified operating threshold (optimised for maximal F1 in the development set).
 2. Assess calibration of predicted probabilities (Brier score, reliability diagram, Hosmer-Lemeshow test).
 3. Determine the lead-time advantage over BISAP score (comparing prediction accuracy at admission vs. 24 h).
 4. Evaluate the secondary pancreatic sepsis risk sub-model against culture-confirmed or clinically diagnosed pancreatic sepsis.
 5. Characterise any systematic performance differences across age, sex, aetiology (gallstone vs. alcohol vs. other), and BMI strata.
 6. Quantify clinical alert burden: proportion of patients flagged as high-risk, false-alert rate.
 7. Assess feasibility of data harmonisation between local EHR variables and PenuX input features.
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3. Study Design

Parameter	Value
Design	Prospective, observational, silent-mode (non-interventional)
Setting	One or more hospital sites with dedicated gastroenterology / acute medicine service
Duration	12 months enrolment + 3 months follow-up
Comparator	Standard clinical severity scores (BISAP, APACHE II at 24 h)
Blinding	Clinicians blinded to PenuX outputs throughout Stage I
Ethics	Full IRB/Helsinki Committee approval required before initiation

Silent-mode rationale: PenuX predictions are logged but not shown to treating clinicians during Stage I. This eliminates feedback loops, preserves standard-of-care as the sole treatment driver, and produces unbiased outcome labels.

4. Study Population

4.1 Inclusion Criteria

- Age ≥ 18 years
- Admission diagnosis of acute pancreatitis meeting the 2012 Revised Atlanta Classification diagnostic criteria (≥ 2 of: characteristic abdominal pain; serum lipase or amylase $\geq 3\times$ upper limit of normal; characteristic imaging findings)
- Admission within 24 hours of symptom onset
- Availability of routine admission blood tests (see Section 6) within 6 hours of presentation
- Informed consent (or waiver of consent approved by IRB for deferred consent in obtunded patients)

4.2 Exclusion Criteria

- Chronic pancreatitis without acute-on-chronic exacerbation
- Post-ERCP pancreatitis (distinct physiology)
- Pre-existing pancreatic malignancy
- Patients admitted directly to ICU from another hospital after ≥ 48 hours of disease course (severity already established)

- Refusal of consent
- Pregnancy (separate physiological reference ranges; deferred to a dedicated sub-study)

4.3 Sample Size

Based on the development-set AUC of 0.877 and assuming a conservative expected external-validation AUC of 0.82 (10% degradation), with H_0 : AUC = 0.70, $\alpha = 0.05$ (one-sided), power = 0.80, and SAP prevalence ~25% in a general hospital setting:

- **Required sample: n = 180 patients** (80 SAP, 100 non-SAP expected)
- Inflated to **n = 220** to account for 15% attrition (incomplete blood panels, loss to follow-up, withdrawal)
- Enrolment target: **~18?20 patients/month** over 12 months at a single site

Sample size calculation performed using the Obuchowski?Mcclish method for AUC comparison (MedCalc v22).

5. Enrolment and Consent Procedure

1. Attending physician or research nurse screens all AP admissions against inclusion/exclusion criteria within 6 hours of presentation.
2. Eligible patients (or their next of kin if patient is obtunded) are approached for written informed consent.
3. An IRB-approved deferred-consent pathway may be used for obtunded patients; full consent obtained at the earliest opportunity.
4. Enrolment logged in REDCap with timestamp, enroller ID, and consent status.

6. Data Collection

6.1 Admission Variables (PenuX Model Inputs)

Collected from the EHR within 6 hours of presentation:

Domain	Variables
Demographics	Age, sex, BMI
Inflammatory	CRP, WBC, Neutrophil %, Neutrophil count, Lymphocyte %, Lymphocyte count, Monocyte %, Monocyte count
Pancreatic enzymes	Amylase (AMY), Lipase (where available)
Liver/biliary	ALT, AST, GGT, ALP, TBIL, DBIL, IBIL, ALB, TP, GLB, A/G Ratio

Domain	Variables
Renal/electrolytes	Creatinine (Cr), Urea, Na ⁺ , K ⁺ , Cl ⁻ , Ca ²⁺
Coagulation	PT, APTT, INR, FIB
Metabolic	Glucose (Glu), TG, LDL-C, CO ₂ -CP
Haematology	RBC, Hb, HCT, MCV, MCH, MCHC, RDW-CV, RDW-SD, PLT, PDW, PCT, P-LCR, MPV
Cardiac	LDH, alpha-HBDH, CK, CK-MB
Other	Eosinophil %, Basophil %, Eosinophil count, Basophil count

6.2 Clinical Outcome Variables

Variable	Timing	Source
SAP label (RAC 2012)	48-72 h post-admission	Clinical assessment by gastroenterologist
BISAP score	Admission, 24 h	Clinical notes
APACHE II score	24 h	ICU chart / clinical notes
Ranson criteria	48 h	Clinical notes
Organ failure (SOFA sub-scores: renal, respiratory, cardiovascular)	Daily x5	EHR
Pancreatic necrosis (CT)	If CT performed (clinical indication)	Radiology report
Infected necrosis / pancreatic sepsis	Confirmed by culture or clinical criteria	Microbiology, clinical notes
ICU admission	Any point during admission	EHR
Length of hospital stay (days)	Discharge	EHR
30-day mortality	30 days	Medical record or phone follow-up
30-day readmission	30 days	Medical record or phone follow-up

6.3 AP Aetiology

Classified by attending physician: gallstone, alcohol, hypertriglyceridaemia, post-ERCP (excluded), idiopathic, other.

7. PenuX System Operation (Silent Mode)

1. Admission lab values are extracted from the EHR (manually or via HL7 interface) and entered into the PenuX web portal or API endpoint.
2. The system outputs:
 - **SAP Severity Probability** (0?1) and binary classification (high-risk / low-risk) at the pre-specified F1-optimal threshold
 - **Pancreatic Sepsis Risk Probability** (0?1)
 - **Risk group label** (low / moderate / high)
 - **Feature attribution** (top-3 contributing variables per patient)
3. All outputs are logged to a secure, access-controlled database. No output is displayed in clinical systems or to treating staff during Stage I.
4. PenuX outputs are linked to patient records via pseudonymised study ID only.

8. Primary and Secondary Endpoints

8.1 Primary Endpoint

- AUC-ROC of PenuX SAP severity probability vs. RAC 2012 severity label at 48 h

8.2 Secondary Endpoints

Endpoint	Metric
Sensitivity at pre-specified threshold	% (95% CI)
Specificity at pre-specified threshold	% (95% CI)
PPV	% (95% CI)
NPV	% (95% CI)
F1 score	(95% CI, bootstrap)
Calibration	Brier score, Hosmer?Lemeshow p-value
Lead time vs. BISAP	AUC at admission vs. AUC at 24 h (DeLong test)
Pancreatic sepsis AUC	AUC-ROC vs. culture-confirmed/clinical sepsis
Subgroup AUC: age ≥ 65	AUC-ROC (95% CI)
Subgroup AUC: gallstone aetiology	AUC-ROC (95% CI)
Alert burden	% patients flagged high-risk per day
Data completeness	% of 59 features available at admission

9. Statistical Analysis Plan

9.1 Primary Analysis

AUC-ROC calculated using the DeLong non-parametric estimator with 95% CI. Pre-specified H0: AUC $\leq$ 0.70 (worse than existing scores). One-sided test, $\alpha = 0.05$.

9.2 Secondary Analyses

- Sensitivity/specificity/PPV/NPV: Wilson score 95% CI
- Calibration: Brier score; HL chi2 statistic (10 groups); reliability plot
- DeLong test for paired AUC comparison (PenuX-admission vs. BISAP-24h)
- Subgroup analyses: Cochran's Q test for heterogeneity across pre-specified strata; no p-value correction (exploratory)
- Missing data: Multiple imputation by chained equations (MICE, $m=10$) for features missing in $<30\%$ of cases; complete-case sensitivity analysis

9.3 Multiple Comparisons

Secondary endpoints are exploratory. No Bonferroni adjustment is applied, but findings are interpreted as hypothesis-generating.

9.4 Software

Python 3.11 (scikit-learn, scipy, statsmodels), R 4.4 (pROC, rms), REDCap for data capture.

10. Data Management and Security

- All patient data stored in REDCap (ISO 27001 certified, hosted within hospital firewall or approved cloud region)
- Pseudonymisation: patient linked to 8-character alphanumeric study ID; linkage key held by site PI only
- PenuX receives only pseudonymised structured data; no free-text, no imaging, no direct identifiers
- Data transfer to PenuX servers (if any) via TLS 1.3 encrypted API; server-side encryption at rest (AES-256)
- Right to access / erasure fulfilled by deleting the study ID from the linkage key
- Data retention: 10 years post-study completion per GCP guidelines; anonymised dataset archived for open science (post-publication)

11. Ethical and Regulatory Considerations

- **IRB/Helsinki approval** required at each site before enrolment commences
- **Informed consent** obtained in writing per Declaration of Helsinki (2013 revision)
- **Non-interventional:** PenuX outputs are never used for clinical decision-making during Stage I; no additional procedures or interventions are imposed on participants
- **CE/FDA:** PenuX is not a CE-marked or FDA-cleared device; Stage I is conducted under the research exemption. Any future clinical use will require appropriate regulatory clearance
- **IEC 62304 lifecycle documentation** is maintained for the PenuX software module
- **Data Protection:** Compliant with GDPR (EU) / Israeli Privacy Protection Law 5741-1981 as applicable

12. Site Requirements

Requirement	Minimum
Annual AP admissions	≥ 80 patients/year
Gastroenterology or acute medicine service	Required
Clinical PI (MD)	Required
Research nurse or data coordinator	Highly recommended
EHR with structured lab data	Required
REDCap access or equivalent	Required
Local IRB/ethics committee	Required
Data protection officer sign-off	Required

13. Timeline

Milestone	Target Month
IRB submission	M1
IRB approval	M2?M3
Site setup, staff training, REDCap configuration	M3?M4
First patient enrolled	M4
50% enrolment (n ? 110)	M9
Last patient enrolled	M16
Last patient 30-day follow-up complete	M17
Database lock	M17
Statistical analysis complete	M18
Manuscript submission	M19

14. Outputs and Dissemination

- Primary manuscript: AUC/calibration/lead-time results submitted to a peer-reviewed gastroenterology or clinical AI journal (target: *Gut*, *Pancreatology*, *The Lancet Digital Health*, or *PLOS Medicine*)
 - Anonymised dataset and analysis code released to GitHub (github.com/netanelcyber/penuX) upon publication
 - Stage II pilot protocol: if primary AUC ≥ 0.80 , initiate design of a prospective silent-to-active pilot with workflow integration and clinical impact assessment
 - Conference abstract: European Pancreatic Club (EPC) or United European Gastroenterology Week (UEGW)
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15. Amendments

Protocol amendments require written approval from the IRB/ethics committee before implementation. Amendments are tracked in the protocol version history and notified to ClinicalTrials.gov within 30 days.

16. References

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6. DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*. 1988;44(3):837?845.
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preparation]

Appendix A ? PenuX Input Feature Mapping

PenuX Feature Name	Common EHR Label	Units	LOINC Code
CRP	C-reactive protein	mg/L	1988-5
WBC	White blood cell count	x109/L	6690-2
Neutrophil %	Neutrophils/100 leukocytes	%	770-8
AMY	Amylase	U/L	1798-8
ALT	Alanine aminotransferase	U/L	1742-6
Cr	Creatinine	?mol/L	2160-0
Ca	Calcium, total	mmol/L	17861-6
ALB	Albumin	g/L	1751-7
GLU	Glucose	mmol/L	2345-7
LDH	Lactate dehydrogenase	U/L	2532-0
PLT	Platelet count	x109/L	777-3
INR	International normalised ratio	ratio	6301-6
TG	Triglycerides	mmol/L	2571-8
Na+	Sodium	mmol/L	2951-2
K+	Potassium	mmol/L	2823-3

Full 59-feature mapping available in the PenuX technical documentation.

Appendix B ? RAC 2012 Severity Classification

Category	Criteria
Mild AP	No organ failure, no local or systemic complications
Moderately Severe AP	Transient organ failure (<48 h) and/or local complications
Severe AP (SAP)	Persistent organ failure (>48 h) ? single or multi-organ

Organ failure assessed by Modified Marshall Scoring System (respiratory, renal, cardiovascular).

This document is a research protocol and does not constitute clinical guidance. PenuX is an investigational system not approved for clinical use.